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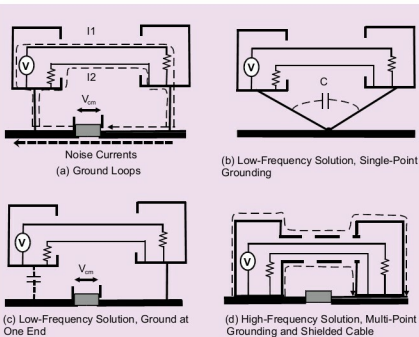


Fig. 20: Reducing ground impedance coupling

grounding is used in situations where systems involve both high-frequency (digital) circuits and low-frequency (analogue) circuits.

Reducing ground impedance coupling

Due to improper grounding and bonding, or due to improper grounding practices, ground impedance tends to increase. This is shown as lumped impedance in the Fig. 20 (a).

EMI currents flowing through this impedance cause voltage drop V_{cm} across it, which forces common-mode EMI currents I1 and I2 through the circuit. The first way is to eliminate this common impedance completely by grounding the modules at a single point as shown in Fig. 20 (b).

The second option is to open the ground loop by grounding only one of the modules as shown in Fig. 20 (c). But these hold good only at low frequencies up to a few tens of kilohertz, where this impedance behaves only as resistance.

At higher frequencies, stray

capacitance (shown as dotted lines) begins to appear between grounding cables or between the module and the ground, causing common-mode currents to circulate again.

Also, at higher frequencies, ground wires themselves start offering high impedance since their inductive reactance and resistance due to skin effect start to increase, causing potential gradients along the wire.

Therefore at high frequencies, efforts should be directed towards reducing ground impedance, which, in turn, can be reduced first by reducing inductance, that is, replacing the wires by a metal sheet (a mounting plate) and then to reduce the value of ground resistance by proper bonding practices.

Also, interconnecting cables should be shielded so that common-mode EMI currents now flow as shown in Fig. 20 (d) on the outside of the shield, reducing common-mode coupling.

This article is an extract from a book by the author. The next part will cover bonding.